

# Molecular structure inference from molecular spectra

good news / bad news

## Phase I – Pretraining

- ✓ Pretrain on SMILES inputs, and:
  - ✓ bias a hidden state towards ChemBERTa-v2,
  - ✓ hijack a hidden state to predict functional groups,
  - ✓ augment with non-canonical SMILES.
- Correctness self-assessment flag – after SMILES add ✓/✗
- Backtracking / inference-time reasoning – provide/teach *undo*
- ✓ Pretrain on the one-atom-one-peak “quasi-NMR”

## Phase II – NMR to SMILES inference training

- ✓ More careful data split (correlations due to USPTO origin)
- ✓ Train on Alberts et al., 2024 [1]
  - Compare vs Alberts et al., 2025 [2]
  - Compare H+C+HSQC vs H+HSQC

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# Phase I – Pretraining

O=C([O-])[O-].[Ca+2][:][\_][o][\_][<=>]C([O-])([O-])=O.[Ca+2] ■

A given SMILES structure

# Phase I – Pretraining

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Unsupervised trigger token (molecular embedding)

# Phase I – Pretraining

$$\text{O}=\text{C}([\text{O}^-])[\text{O}^-].[\text{Ca}+2][:] \text{[red box]} [\circ][\text{—}][\rightleftharpoons] \text{C}([\text{O}^-])([\text{O}^-])=\text{O}.[\text{Ca}+2] \blacksquare$$

Supervised hidden state (molecular embedding)

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$$\text{O}=\text{C}([\text{O}^-])[\text{O}^-].[\text{Ca}+2][:] [\_][\text{O}][\_][\rightleftharpoons]\text{C}([\text{O}^-])([\text{O}^-])=\text{O}.[\text{Ca}+2] \blacksquare$$

Unsupervised trigger token (functional groups)

# Phase I – Pretraining



Supervised hidden state (functional groups)



# Phase I – Pretraining

O=C([O-])[O-].[Ca+2][:][\_][o][\_][<=>]C([O-])([O-])=O.[Ca+2] ■

Unsupervised trigger token (alt SMILES)

# Phase I – Pretraining

O=C([O-])[O-].[Ca+2][:][\_][o][\_][<=>C([O-])([O-])=O.[Ca+2] █

Distribution-supervised alternative SMILES

# Phase I – Pretraining

$\text{O}=\text{C}([\text{O}^-])[\text{O}^-].[\text{Ca}+2][:] [\_][\circ][\_][\rightleftharpoons]\text{C}([\text{O}^-])([\text{O}^-])=\text{O}.[\text{Ca}+2]$  

Supervised “end of sequence”

# Phase I – Pretraining: losses definitions

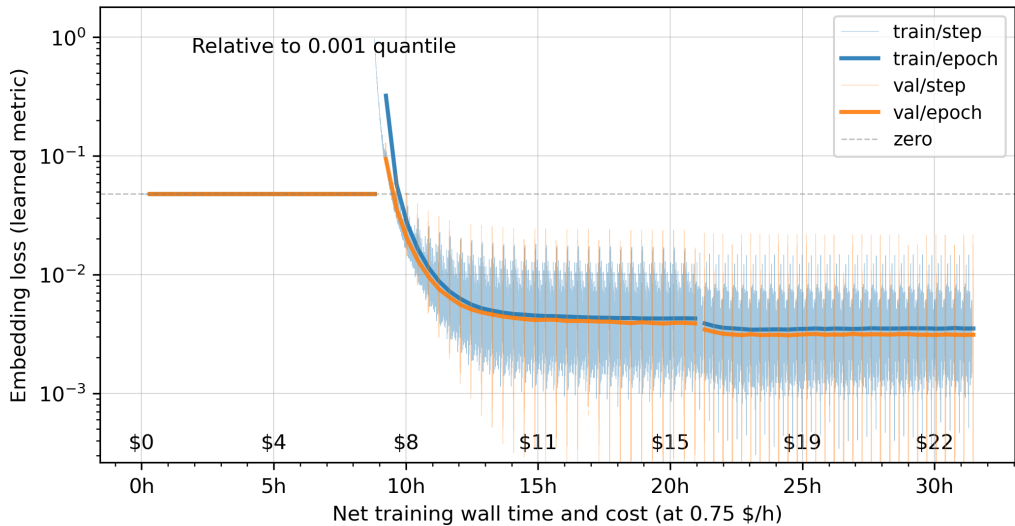
- $\frac{1}{n}(\frac{1}{2}\|T\mathbf{e}\|_2^2 - \log \det T + \frac{n}{2} \log 2\pi)$  – embedding ( $n = 384$ )
- $\frac{1}{n}(\|T\mathbf{e}\|_1 - \log \det T + n \log 2)$  – functional groups ( $n = 289$ )
- KL divergence, position-averaged – next-token distribution
- Causal LM =  $-\log p(\text{eos} \mid \text{prefix})$  – end-of-sequence token

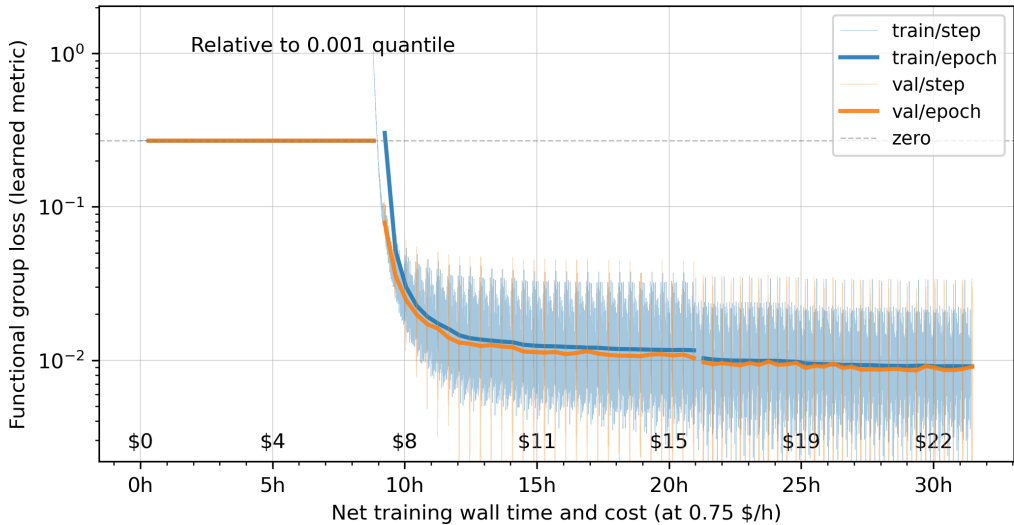
combined à la Kendall et al. [3]:

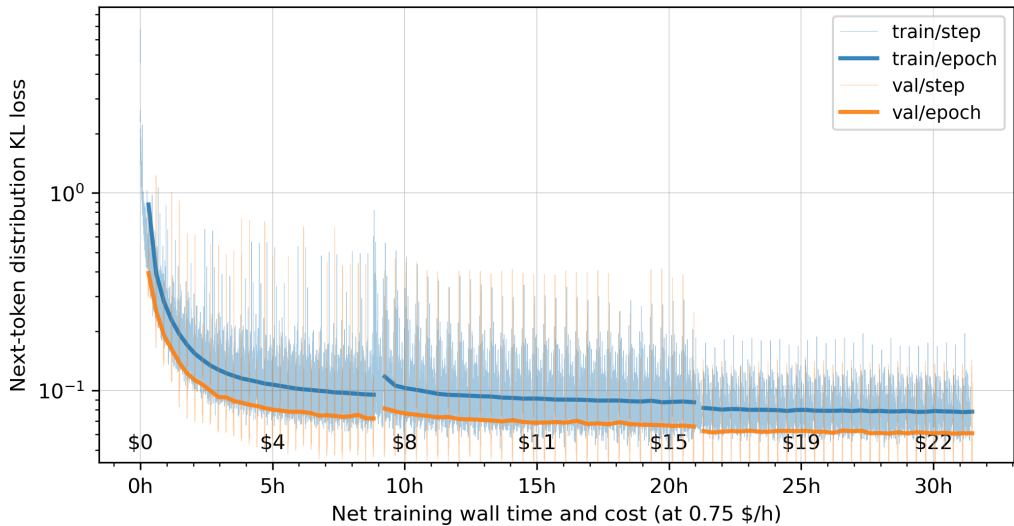
$$\text{minimize} \quad \sum_t \left( \frac{1}{\sigma_t^2} \text{softplus}(\text{loss}_t) + \log \sigma_t^2 \right)$$

with a trainable  $\sigma_t$  for each task  $t$ .

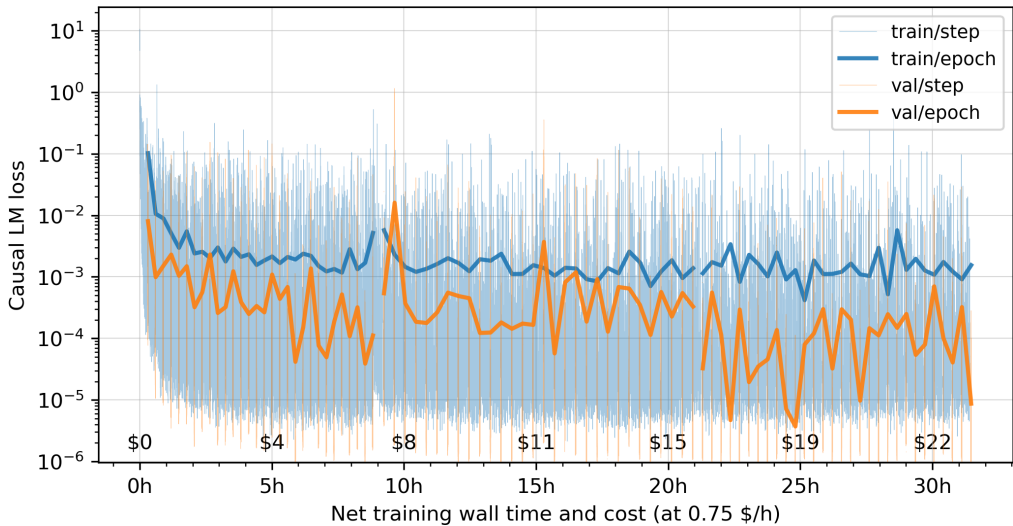
Losses  
during training











# Intuition for the KL loss

Suppose

$$P \sim \text{uniform over a set} \quad \text{and} \quad Q \sim (1 - q)P + qP^c.$$

Then

$$\text{KL}(P \mid Q) = -\ln(1 - q) \approx q$$

How good are the predictions?

“Flag plot” of 10-out-of-20 beam search on 200 test samples



is\_exact



is\_isomer



is\_sum\_ok



is\_majority



char\_len



score

low → high by: is\_exact



is\_exact

A heatmap visualization for the variable 'is\_exact'. The plot shows a dense pattern of blue and red vertical stripes across the entire width, indicating high variability in the data.



is\_isomer

A heatmap visualization for the variable 'is\_isomer'. The plot shows a dense pattern of blue and red vertical stripes across the entire width, indicating high variability in the data.



is\_sum\_ok

A heatmap visualization for the variable 'is\_sum\_ok'. The plot shows a dense pattern of blue and red vertical stripes across the entire width, indicating high variability in the data.



is\_majority

A heatmap visualization for the variable 'is\_majority'. The plot shows a dense pattern of blue and red vertical stripes across the entire width, indicating high variability in the data.



char\_len

A heatmap visualization for the variable 'char\_len'. The plot shows a dense pattern of blue and red vertical stripes across the entire width, indicating high variability in the data.



score

A heatmap visualization for the variable 'score'. The plot shows a dense pattern of blue and red vertical stripes across the entire width, indicating high variability in the data.

low → high by: is\_isomer



is\_exact

A horizontal bar chart for the variable 'is\_exact'. The bar is predominantly red, indicating high values, with some blue segments at the beginning and end, indicating low values.



is\_isomer

A horizontal bar chart for the variable 'is\_isomer'. The bar is predominantly red, indicating high values, with some blue segments at the beginning and end, indicating low values.



is\_sum\_ok

A horizontal bar chart for the variable 'is\_sum\_ok'. The bar is predominantly red, indicating high values, with some blue segments at the beginning and end, indicating low values.



is\_majority

A horizontal bar chart for the variable 'is\_majority'. The bar is predominantly red, indicating high values, with some blue segments at the beginning and end, indicating low values.



char\_len

A horizontal bar chart for the variable 'char\_len'. The bar is predominantly blue, indicating low values, with some red segments at the beginning and end, indicating high values.



score

A horizontal bar chart for the variable 'score'. The bar is predominantly red, indicating high values, with some blue segments at the beginning and end, indicating low values.

low → high by: is\_sum\_ok



A horizontal bar chart for the variable 'is\_exact'. The bar is predominantly red, with a small blue segment at the left end. A vertical white line is positioned near the right end of the bar.

is\_exact



A horizontal bar chart for the variable 'is\_isomer'. The bar is predominantly red, with a small blue segment at the left end. A vertical white line is positioned near the right end of the bar.

is\_isomer



A horizontal bar chart for the variable 'is\_sum\_ok'. The bar is predominantly red, with a small blue segment at the left end. A vertical white line is positioned near the right end of the bar.

is\_sum\_ok



A horizontal bar chart for the variable 'is\_majority'. The bar is predominantly red, with a small blue segment at the left end. A vertical white line is positioned near the right end of the bar.

is\_majority



A horizontal bar chart for the variable 'char\_len'. The bar is filled with a dense, vertical pattern of red and blue stripes.

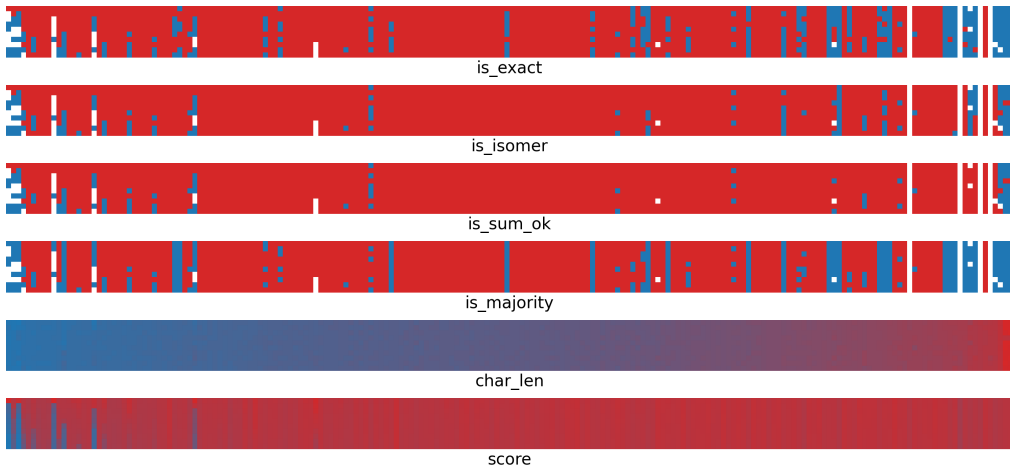
char\_len



A horizontal bar chart for the variable 'score'. The bar is filled with a dense, vertical pattern of red and blue stripes.

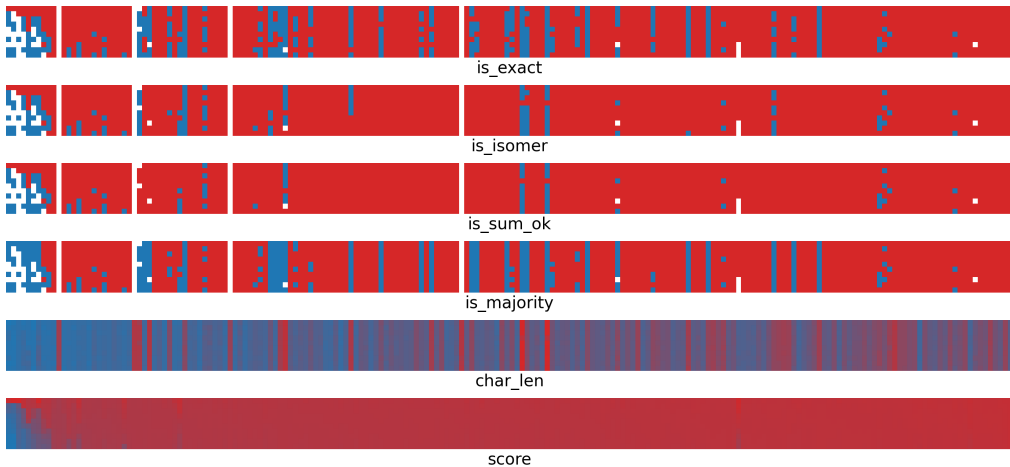
score

low → high by: is\_majority



low → high by: `char_len`





low → high by: score

## Phase II – NMR to SMILES with next-token distribution

F|C13H10N2O6

Q|2.0:5.61-5.72@89.9|1.0:8.12-8.23@126|1.0:7.02-7.13@115

C|10@163|10@142|34@126|34@115|12@90

H|4:8.15-8.2:m|4:7.06-7.1:m|2:5.66-5.68:s

[:]\_[]\_[]O=[N+]([O-])c1ccc(OCOc2ccc([N+](=O)[O-])cc2)cc1

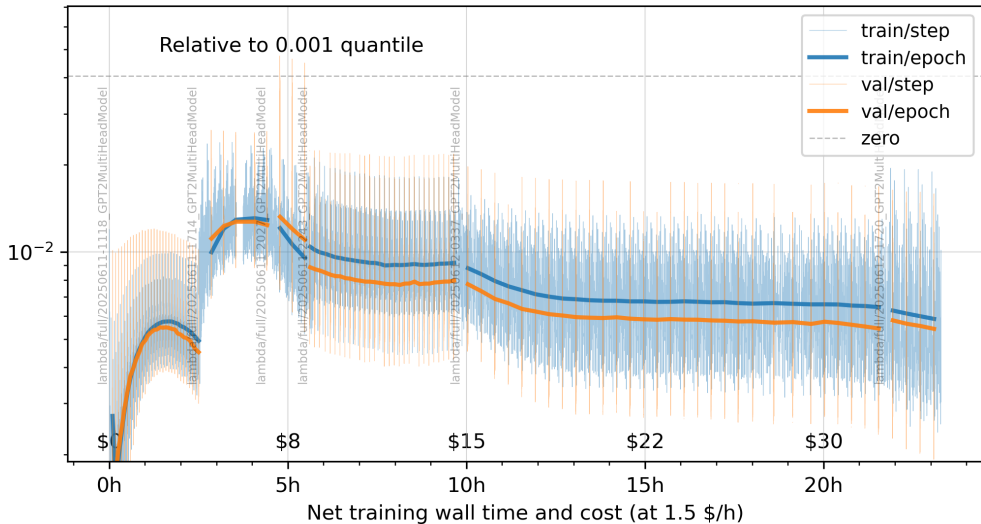
## Phase II – NMR to SMILES with next-token distribution

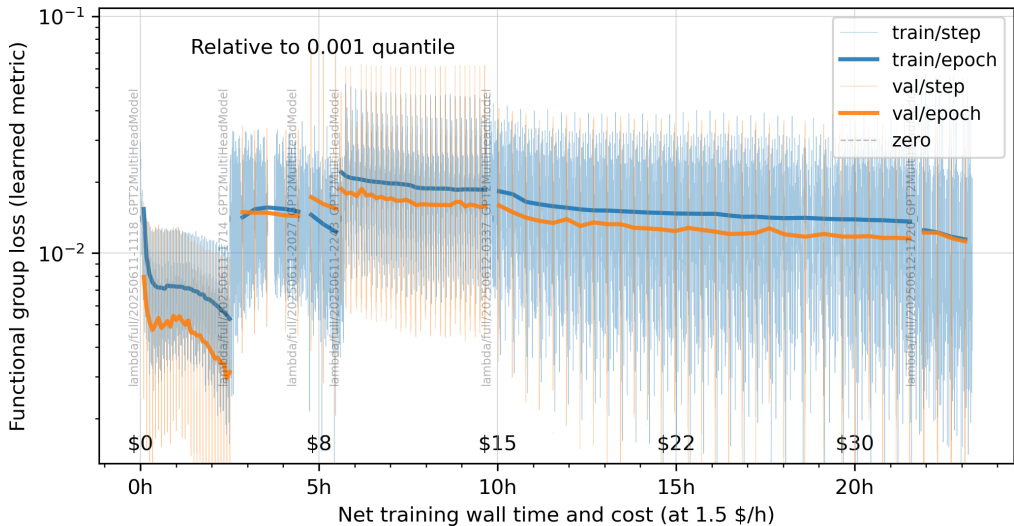
- Training resumed from the model pretrained in Phase I
- 3 runs of quasi-NMR on up to  $\sim 600$ k PubChem SMILES
- 3 runs of MestReNova NMR on up to  $\sim 600$ k “Alberts” SMILES

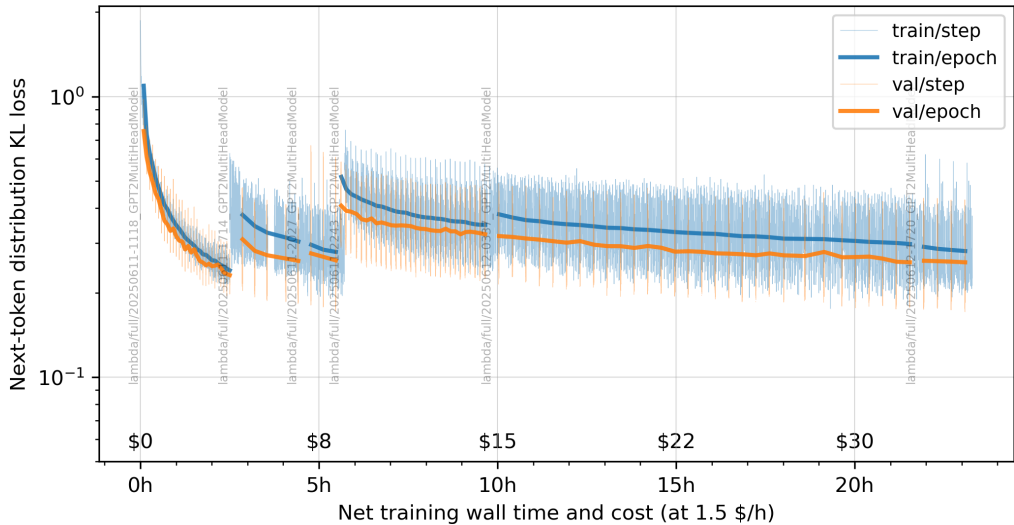
Losses  
during training

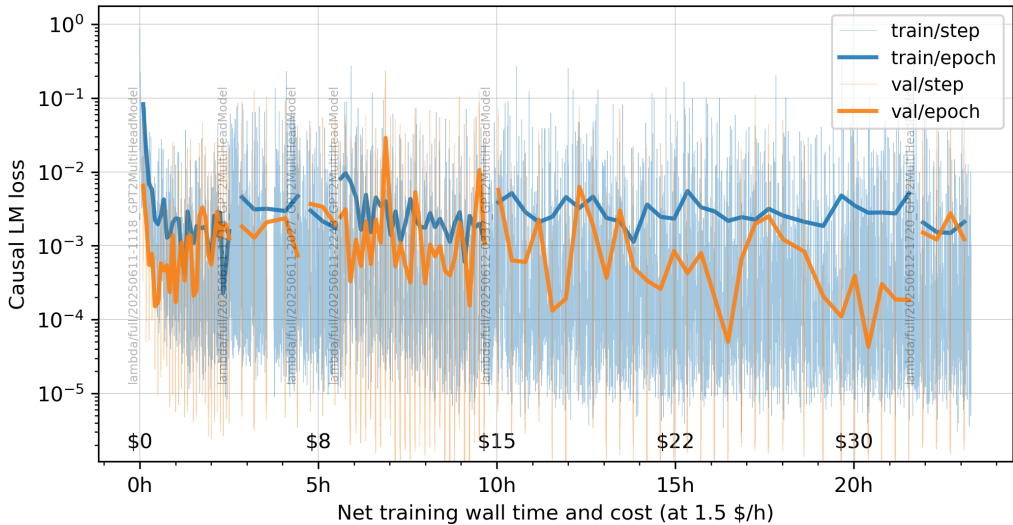
Embedding loss (learned metric)

Relative to 0.001 quantile











It's learning but slowly

- Faster on quasi-NMR
- KL loss is  $\sim 30\%$
- AdaBelief: scales parameter updates by predictability of gradient

Things to try

- Learning rate
- Penalize out-of-distribution tokens more
- Drop / re-prioritize tasks
- Ditch all pretraining

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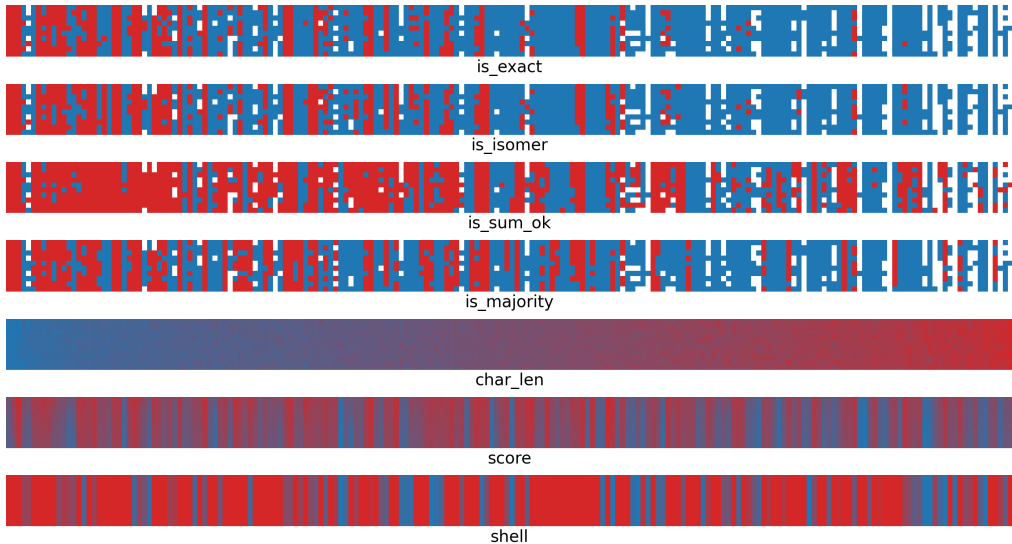


score



shell

low → high by: is\_majority



low → high by: char\_len



is\_exact



is\_isomer



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shell

low → high by: score





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char\_len



score



shell

low → high by: shell

thx.

# References I

- [1] Marvin Alberts, Oliver Schilter, Federico Zipoli, Nina Hartrampf, and Teodoro Laino. “Unraveling Molecular Structure: A Multimodal Spectroscopic Dataset for Chemistry”. In: *NeurIPS 2024 Datasets and Benchmarks Track*. 2024. [url].
- [2] Marvin Alberts, Nina Hartrampf, and Teodoro Laino. *Automated structure elucidation at human-level accuracy via a multimodal multitask language model*. 2025. [url].
- [3] Alex Kendall, Yarin Gal, and Roberto Cipolla. “Multi-Task Learning Using Uncertainty to Weigh Losses for Scene Geometry and Semantics”. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (2018). [url].

## Appendix

# GPT-2 reference implementation<sup>1</sup>

**Input:**  $x \in \mathbb{N}^{B \times T}$  (token indices)

Token embedding:  $\mathbf{E}_{\text{tok}} = \text{Embedding}(x) \in \mathbb{R}^{B \times T \times d}$

Position embedding:  $\mathbf{E}_{\text{pos}} = \text{Embedding}[1 : T]$

Combined:  $\mathbf{H}^{(0)} = \text{Dropout}(\mathbf{E}_{\text{tok}} + \mathbf{E}_{\text{pos}})$

**For each block**  $\ell = 1 \dots L$ :

$$\mathbf{X} = \mathbf{H}^{(\ell-1)}$$

$$\mathbf{X} \leftarrow \mathbf{X} + \text{CausalSelfAttn}(\text{LayerNorm}(\mathbf{X}))$$

$$\mathbf{X} \leftarrow \mathbf{X} + \text{MLP}(\text{LayerNorm}(\mathbf{X}))$$

$$\mathbf{H}^{(\ell)} = \mathbf{X}$$

Final:  $\mathbf{H}_{\text{final}} = \text{LayerNorm}(\mathbf{H}^{(L)})$

Logits:  $\hat{y} = \mathbf{H}_{\text{final}} \cdot \mathbf{E}_{\text{tok}}^{\top}$  (weight tying)

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<sup>1</sup>Based on <https://github.com/karpathy/nanoGPT/blob/master/model.py>

